

WORKSHEET - NAPLAN Year 9

Number and Algebra

Fractions, Decimals and Percentages (9)

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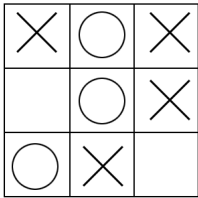
Exam Equivalent Time: 8.75 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. Number and Algebra, NAP-61527

The image below shows a game of noughts and crosses in progress.



What fraction of the available boxes have been filled in?

- ☐ $\frac{1}{2}$ ☐ $\frac{4}{7}$ ☐ $\frac{5}{7}$ ☐ $\frac{7}{9}$

2. Number and Algebra, NAP-02657

James weighed 80 kg when he was 18 years old.

On his 21st birthday, James weighs 5% more than when he was 18.

How much does he weigh now?

- ☐ 76 kg ☐ 84 kg ☐ 85 kg ☐ 120 kg

3. Number and Algebra, NAP-20265

A large birdcage contains 8 canaries and 13 parakeets.

About what percentage of the birds are canaries?

- ☐ 8% ☐ 38% ☐ 58% ☐ 62%

4. Number and Algebra, NAP-02685

A christmas pudding recipe uses 3 cups of sugar for every 4 cups of sultanas.

Select the correct combination of sugar and sultanas for this recipe.

- ☐ $\frac{1}{3}$ cup of sugar, $\frac{1}{4}$ cup of sultanas
☐ 1 cup of sugar, 1.5 cups of sultanas
☐ 1.5 cups of sugar, 2 cups of sultanas
☐ 4 cups of sugar, 5 cups of sultanas

5. Number and Algebra, NAP-79164

Alison's petrol tank was empty.

She then spent \$55 filling her tank up to half way at a cost of \$1.55 per litre.

Approximately how much petrol can Alison's petrol tank hold when it is full?

- ☐ 35 litres ☐ 65 litres ☐ 71 litres ☐ 85 litres

6. Number and Algebra, NAP-38067

Which of these percentages is closest in value to $\frac{5}{9}$?

- ☐ 54% ☐ 55% ☐ 56% ☐ 57%

7. Number and Algebra, NAP-02720

A marine conservationist records the number of eggs laid by 3 turtles and the number of those eggs that hatched.

Her results are recorded in the table below:

Turtle	Eggs laid	Percentage that hatched
1	300	25%
2	120	75%
3	80	50%

What percentage of the eggs from all 3 turtles hatched?

 %

Worked Solutions

1. Number and Algebra, NAP-61527

$$\begin{aligned}\text{Fraction} &= \frac{\text{spaces filled}}{\text{total spaces}} \\ &= \frac{7}{9}\end{aligned}$$

2. Number and Algebra, NAP-02657

$$\begin{aligned}80 + (80 \times 5\%) &= 80 + 4 \\ &= 84 \text{ kg}\end{aligned}$$

3. Number and Algebra, NAP-20265

$$\frac{8}{8 + 13} = \frac{8}{21} = 0.380\dots$$

$\therefore$ Approx 38% are canaries.

4. Number and Algebra, NAP-02685

$$\begin{aligned}\text{Ratio} &= 3:4 \\ &= 1.5:2\end{aligned}$$

$\therefore$ 1.5 cups of sugar, 2 cups of sultanas is in the correct ratio.

5. Number and Algebra, NAP-79164

$$\begin{aligned}\text{Volume of half tank} &= \frac{55}{1.55} \\ &= 35.48 \dots\end{aligned}$$

$$\begin{aligned}\therefore \text{Volume of full tank} \\ &= 2 \times 35.48 \dots \\ &= 70.96 \dots \\ &= 71 \text{ litres}\end{aligned}$$

6. Number and Algebra, NAP-38067

$$\begin{aligned}\frac{5}{9} &= 55.55 \dots \% \\ &= 56\% \text{ (nearest \%)}\end{aligned}$$

7. Number and Algebra, NAP-02720

$$\begin{aligned}\text{Total turtle eggs hatched} \\ &= (25\% \times 300) + (75\% \times 120) + (50\% \times 80) \\ &= 75 + 90 + 40 \\ &= 205\end{aligned}$$

$$\begin{aligned}\text{Total eggs laid} &= 300 + 120 + 80 \\ &= 500\end{aligned}$$

$$\begin{aligned}\therefore \text{Percentage of eggs that hatched} \\ &= \frac{205}{500} \times 100 \\ &= 41\%\end{aligned}$$